ORIGINAL ARTICLE

Trastuzumab Deruxtecan plus Pertuzumab for HER2-Positive Metastatic Breast Cancer

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ABSTRACT

BACKGROUND

Trastuzumab deruxtecan has shown efficacy in patients with previously treated human epidermal growth factor receptor 2 (HER2)–positive advanced or metastatic breast cancer. The efficacy and safety of trastuzumab deruxtecan in patients with no previous therapy for HER2-positive advanced or metastatic breast cancer are unclear.

METHODS

We conducted a phase 3 trial involving patients with HER2-positive advanced or metastatic breast cancer and no previous chemotherapy or HER2-directed therapy for metastatic disease. Patients were randomly assigned in a 1:1:1 ratio to receive trastuzumab deruxtecan plus pertuzumab; trastuzumab deruxtecan plus placebo; or a taxane, trastuzumab, and pertuzumab (THP). The primary end point was progression-free survival as assessed by blinded independent central review. Secondary end points included objective response, duration of response, and safety.

RESULTS

For this prespecified interim analysis, data for trastuzumab deruxtecan plus pertuzumab and for THP are reported; data for trastuzumab deruxtecan plus placebo remain blinded until the final analysis of progression-free survival. At the data-cutoff date (February 26, 2025), the median progression-free survival was 40.7 months with trastuzumab deruxtecan plus pertuzumab (383 patients) and 26.9 months with THP (387 patients) (hazard ratio for progression or death, 0.56; 95% confidence interval [CI], 0.44 to 0.71; P<0.00001 [P-value boundary for superiority, 0.00043]). The incidence of a confirmed response was 85.1% with trastuzumab deruxtecan plus pertuzumab and 78.6% with THP (complete responses in 15.1% and 8.5%, respectively), with a median duration of response of 39.2 months and 26.4 months. Safety was consistent with the known profiles of the individual treatments. The incidence of grade 3 or higher adverse events was 63.5% with trastuzumab deruxtecan plus pertuzumab and 62.3% with THP; the most common were neutropenia, hypokalemia, and anemia with trastuzumab deruxtecan plus pertuzumab and neutropenia, leukopenia, and diarrhea with THP. Adjudicated drug-related interstitial lung disease or pneumonitis occurred in 12.1% of patients receiving trastuzumab deruxtecan plus pertuzumab (grade 1 or 2 in 44 patients and grade 5 [death] in 2 patients) and in 1.0% of those receiving THP (all grade 1 or 2).

CONCLUSIONS

Trastuzumab deruxtecan plus pertuzumab led to a significantly lower risk of progression or death than THP when used as first-line treatment for HER2-positive advanced or metastatic breast cancer, with no new safety signals. (Funded by AstraZeneca and Daiichi Sankyo; DESTINY-Breast09 ClinicalTrials.gov number, NCT04784715.)

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*A list of the principal investigators in the DESTINY-Breast09 trial is provided in the Supplementary Appendix, available at NEJM.org.

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Human epidermal growth factor receptor (HER) 2–positive breast cancer (immunohistochemistry 3+ or in situ hybridization–positive)[1] accounts for approximately 20% of breast cancers.[2,3] HER2-targeted therapies have improved outcomes that had previously been poor,[4] but many patients do not receive a diagnosis until their disease has advanced or metastasized,[5] while others go on to have metastases despite initial treatment.[4] Standard first-line therapy for HER2-positive advanced or metastatic breast cancer is a taxane plus the HER2-directed monoclonal antibodies trastuzumab and pertuzumab (the three-drug regimen is known as THP).[6-8] The use of THP is based on the primary analysis of the 2012 phase 3 Clinical Evaluation of Pertuzumab and Trastuzumab (CLEOPATRA) trial, which established that, added to chemotherapy, dual HER2 blockade resulted in significantly longer progression-free survival than trastuzumab alone (median, 18.5 vs. 12.4 months; P<0.001).[9,10]

Antibody–drug conjugates have improved therapeutic options for HER2-expressing cancers and are recommended when HER2-positive advanced or metastatic breast cancer has progressed after first-line treatment.[6-8] Trastuzumab deruxtecan (also known as T-DXd) comprises a humanized immunoglobulin G1 monoclonal antibody (trastuzumab, which inhibits HER2 homodimerization) joined to a potent topoisomerase I inhibitor chemotherapy payload by a tetrapeptide-based cleavable linker.[11-13] In the phase 3 DESTINY-Breast03 trial, trastuzumab deruxtecan monotherapy improved outcomes as compared with the (then) standard of care, trastuzumab emtansine (T-DM1), in the second-line metastatic context.[14-16] Trastuzumab deruxtecan was therefore approved after failure of a previous HER2-directed regimen[17,18] and is the second-line standard of care.[6,7]

Pertuzumab, which prevents heterodimerization of HER2 and HER3, and trastuzumab show complementary inhibitory effects on tumor-cell proliferation and survival,[19] and this synergy may also apply to trastuzumab deruxtecan plus pertuzumab. In the phase 1b–2 DESTINY-Breast07 trial, initial results indicated promising efficacy and no new safety concerns with trastuzumab deruxtecan alone or in combination with pertuzumab for first-line treatment of HER2-positive metastatic breast cancer.[20] The phase 3 DESTINY-Breast09 trial was conceived to evaluate the efficacy and safety of trastuzumab deruxtecan monotherapy and trastuzumab deruxtecan plus pertuzumab in this context. Here we report results from a prespecified interim analysis.

METHODS

TRIAL DESIGN AND TREATMENT

This phase 3, multinational, randomized trial compared trastuzumab deruxtecan plus pertuzumab or placebo (blinded for pertuzumab or placebo in the experimental groups) with open-label THP for first-line treatment in patients with centrally confirmed advanced or metastatic HER2-positive breast cancer. Patients had received no previous chemotherapy or HER2-directed therapy for metastatic disease. Patients who received adjuvant or neoadjuvant chemotherapy with a disease-free interval of more than 6 months (183 days) between the completion of systemic anticancer therapy and a diagnosis of advanced or metastatic disease were eligible. One previous line of endocrine monotherapy was permitted in the context of metastatic disease.

Patients were randomly assigned in a 1:1:1 ratio to receive trastuzumab deruxtecan plus pertuzumab; trastuzumab deruxtecan plus placebo; or THP (paclitaxel or docetaxel [per physician choice] plus trastuzumab and pertuzumab). If trastuzumab deruxtecan was discontinued owing to adverse events (except grade ≥2 interstitial lung disease, after which no treatment with trastuzumab deruxtecan or trastuzumab was permitted), patients could remain in the trial and receive trastuzumab plus pertuzumab or trastuzumab plus placebo (according to randomized group). THP was administered according to local institutional standards; patients received a taxane for a minimum of six cycles or until the occurrence of unacceptable toxic effects, then continued with trastuzumab plus pertuzumab. Patients with hormone receptor–positive disease could receive concurrent endocrine therapy (aromatase inhibitor or tamoxifen) after taxane discontinuation or six cycles of trastuzumab deruxtecan.

Randomization was performed with the use of interactive response technology, and patients were stratified according to newly diagnosed or recurrent disease, hormone-receptor status, and the presence or absence of *PIK3CA* mutation (centrally tested before enrollment). Efficacy analyses included all the patients who underwent randomization and were performed according to assigned

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treatment group (regardless of actual treatment received). Additional details are provided in the Methods section in the Supplementary Appendix, available with the full text of this article at NEJM.org.

TRIAL OVERSIGHT

The International Council for Harmonisation Good Clinical Practice guidelines, the principles of the Declaration of Helsinki, and local regulations governing the conduct of clinical research were followed. The trial was sponsored by AstraZeneca and Daiichi Sankyo. AstraZeneca was responsible for the design (with Daiichi Sankyo), which was approved by the institutional review board or ethics committee at each investigational site before the start of the trial. Patients provided written informed consent before participating, and an independent data-monitoring committee monitored patient safety.

The first draft of the manuscript was prepared by a professional writer under the direction of the authors and representatives of the sponsors, with funding provided by AstraZeneca. All drafts were reviewed by the authors, who approved the final submitted version and vouch for the completeness and accuracy of the data and for the adherence of the trial to the protocol and statistical analysis plan (available at NEJM.org).

END POINTS

The primary end point was progression-free survival as assessed by blinded independent central review, with disease progression defined according to Response Evaluation Criteria in Solid Tumors, version 1.1. Secondary end points included progression-free survival as assessed by investigators, objective response, duration of response, and safety.

SAFETY

Safety analyses included all the patients who received at least one dose of a trial drug. We collected data on adverse events that emerged or worsened in severity between the initiation of trial treatment and the safety follow-up visit at day 40 (with a window of ±7 days). Adverse events were coded with the use of the preferred terms of the *Medical Dictionary for Regulatory Activities*, version 27.1, and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.0. The relatedness of adverse events to trial treatment was determined by the investigators. Among adverse events of special interest, cases of potential interstitial lung disease or pneumonitis were evaluated by an independent adjudication committee and managed as described previously.[21,22] Additional details are provided in the Methods section in the Supplementary Appendix.

STATISTICAL ANALYSIS

The planned sample size was 1134 patients. The current interim analysis was planned when approximately 399 events of disease progression or death had occurred across all three treatment groups, with at least 277 events between each trastuzumab deruxtecan–based group and the THP (control) group and an identical, stringent, prespecified P-value boundary for superiority of 0.00043 for both comparisons. Progression-free survival was analyzed with the use of a stratified log-rank test, with hazard ratios and two-sided 95% confidence intervals estimated from a stratified Cox proportional-hazards model. Beyond the analysis of progression-free survival, corresponding confidence intervals for reported hazard ratios were not adjusted for multiplicity. Additional details are provided in the Methods section in the Supplementary Appendix.

RESULTS

PATIENTS

Between April 26, 2021, and October 26, 2023, a total of 1703 patients from 279 sites were screened, and 1157 were enrolled and underwent randomization (Fig. S1 in the Supplementary Appendix). The interim efficacy boundary of superiority ($P<0.00043$) was crossed for the comparison between trastuzumab deruxtecan plus pertuzumab and THP, whereas data for trastuzumab deruxtecan plus placebo will remain blinded until the final analysis of progression-free survival.

The demographic and disease characteristics of the patients at baseline were balanced between those assigned to receive trastuzumab deruxtecan plus pertuzumab (383 patients) and those assigned to receive THP (387 patients) (Table 1). Across the two groups, the median age of the patients was 54 years (range, 20 to 85); 400 patients (51.9%) had newly diagnosed disease, 416 (54.0%) had hormone receptor–positive status, and 237 (30.8%) had *PIK3CA* mutations. A total of 335 patients

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(43.5%) received previous adjuvant or neoadjuvant therapy (chemotherapy, 311 patients [40.4%]; trastuzumab, 218 [28.3%]; and pertuzumab, 55 [7.1%]). In the context of metastatic disease, 10 patients (1.3%) had received one line of endocrine therapy. Black patients were underrepresented.

At the data-cutoff date (February 26, 2025), 174 of 380 patients (45.8%) who had received trastuzumab deruxtecan plus pertuzumab and 128 of 383 patients (33.4%) who had received THP were receiving ongoing treatment. The median duration of follow-up was 29.2 months (range, 0.0 to 44.7). Reasons for discontinuing trastuzumab deruxtecan or trastuzumab included progressive disease (21.3% for trastuzumab deruxtecan and 52.5% for trastuzumab) and adverse events (20.8% for trastuzumab deruxtecan and 3.4% for trastuzumab); 33 of 380 patients (8.7%) who had received trastuzumab deruxtecan plus pertuzumab discontinued trastuzumab deruxtecan for reasons other than disease progression and received trastuzumab plus pertuzumab. Among

Table 1. Characteristics of the Patients at Baseline (Full Analysis Population).*

Characteristic	Trastuzumab Deruxtecan plus Pertuzumab (N=383)	THP (N=387)
Median age (range) — yr	54 (27–85)	54 (20–81)
Female sex — no. (%)	383 (100)	387 (100)
Race — no. (%)†		
Asian	190 (49.6)	196 (50.6)
White	138 (36.0)	146 (37.7)
Black	12 (3.1)	5 (1.3)
Other	19 (5.0)	21 (5.4)
Not reported	24 (6.3)	19 (4.9)
Geographic region — no. (%)		
Asia	188 (49.1)	191 (49.4)
Western Europe and North America	87 (22.7)	78 (20.2)
Rest of the world	108 (28.2)	118 (30.5)
ECOG performance-status score — no. (%)‡		
0	256 (66.8)	246 (63.6)
1	127 (33.2)	141 (36.4)
Hormone-receptor status — no. (%)		
Positive	207 (54.0)	209 (54.0)
Negative	176 (46.0)	178 (46.0)
HER2 status on IHC analysis — no. (%)		
2+	56 (14.6)	69 (17.8)
3+	318 (83.0)	315 (81.4)
HER2-low status: IHC 2+ and ISH-negative	0	0
HER2 status on ISH — no. (%)		
Positive	364 (95.0)	373 (96.4)
Negative	1 (0.3)	0
Data not available	18 (4.7)	14 (3.6)
Newly diagnosed disease — no. (%)	200 (52.2)	200 (51.7)
CNS metastasis — no. (%)§	25 (6.5)	22 (5.7)
Visceral metastasis — no. (%)	281 (73.4)	268 (69.3)
PIK3CA mutation — no. (%)	116 (30.3)	121 (31.3)
Any metastatic site — no. (%)	379 (99.0)	383 (99.0)

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Table 1. (Continued.)

Characteristic	Trastuzumab Deruxtecan plus Pertuzumab (N=383)	THP (N=387)
Previous adjuvant or neoadjuvant therapy — no. (%)¶	166 (43.3)	169 (43.7)
Chemotherapy	159 (41.5)	152 (39.3)
Taxane	134 (35.0)	136 (35.1)
Anthracycline	108 (28.2)	111 (28.7)
Cyclophosphamide	111 (29.0)	108 (27.9)
Carboplatin	25 (6.5)	22 (5.7)
Other	36 (9.4)	29 (7.5)
Endocrine therapy	74 (19.3)	85 (22.0)
Tamoxifen	48 (12.5)	52 (13.4)
Fulvestrant	0	2 (0.5)
Anastrozole or letrozole	31 (8.1)	39 (10.1)
Exemestane	8 (2.1)	4 (1.0)
Gonadotropin-releasing hormone analogues	8 (2.1)	10 (2.6)
Other	1 (0.3)	5 (1.3)
Targeted therapy	112 (29.2)	108 (27.9)
CDK4/6 inhibitor	0	1 (0.3)
Trastuzumab	110 (28.7)	108 (27.9)
Pertuzumab	31 (8.1)	24 (6.2)
Pyrotinib	1 (0.3)	1 (0.3)
Trastuzumab emtansine	3 (0.8)	4 (1.0)
Hormonal therapy in first-line advanced or metastatic context — no. (%)	5 (1.3)	5 (1.3)

* The full analysis population included the patients who underwent randomization. Percentages may not total 100% because of rounding. CDK4/6 denotes cyclin-dependent kinases 4 and 6, HER2 human epidermal growth factor receptor 2, IHC immunohistochemical, ISH in situ hybridization, and THP a taxane, trastuzumab, and pertuzumab.
† Race was reported by the patients.
‡ Eastern Cooperative Oncology Group (ECOG) performance-status scores are assessed on a 5-point scale, with higher scores indicating greater disability.
§ Patients with clinically active central nervous system (CNS) metastasis (defined as disease that was untreated, was causing symptoms, or warranted therapy with glucocorticoids or anticonvulsants to control associated symptoms) were excluded.
¶ Data shown are for any treatment that was stopped before the start of trial treatment.

patients with hormone receptor–positive disease, concurrent endocrine therapy was received by 28 of 207 patients (13.5%) in the group assigned to receive trastuzumab deruxtecan plus pertuzumab and by 80 of 209 patients (38.3%) in the THP group.

EFFICACY

Progression-free Survival

At this interim analysis (37.7% maturity for progression-free survival across both groups analyzed), the median progression-free survival as assessed by blinded independent central review was 40.7 months (95% confidence interval [CI], 36.5 to not calculable) with trastuzumab deruxtecan plus pertuzumab and 26.9 months (95% CI, 21.8 to not calculable) with THP (hazard ratio for disease progression or death, 0.56; 95% CI, 0.44 to 0.71; P<0.00001 [P-value boundary for superiority, 0.00043]) (Fig. 1A); progression-free survival at 24 months was 70.1% (95% CI, 64.8 to 74.8) with trastuzumab deruxtecan plus pertuzumab and 52.1% (95% CI, 46.4 to 57.5) with THP. The median progression-free survival as assessed by investigators was 40.7 months (95% CI, 36.5 to not calculable) with trastuzumab deruxtecan plus

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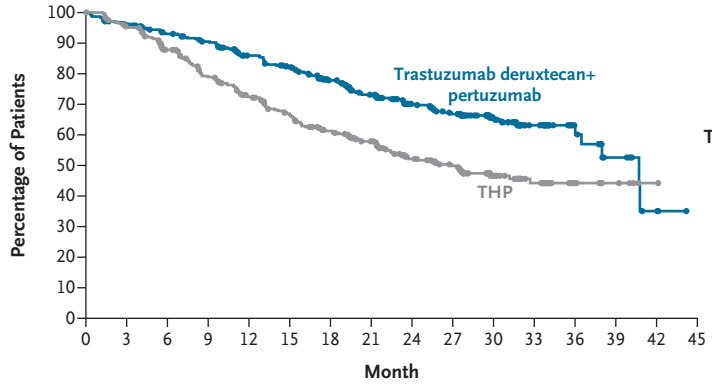
A Progression-free Survival



	No. of Patients	Median Progression-free Survival (95% CI)
		mo
Trastuzumab Deruxtecan+ Pertuzumab	383	40.7 (36.5–NC)
THP	387	26.9 (21.8–NC)

Hazard ratio for disease progression or death, 0.56 (95% CI, 0.44–0.71)
P<0.00001 (prespecified P-value boundary for superiority, 0.00043)

No. at Risk

Trastuzumab deruxtecan+ pertuzumab	383	358	335	321	293	275	242	208	175	153	82	49	21	10	3	0
THP	387	353	312	273	241	215	187	160	124	106	51	32	12	5	1	0

B Subgroup Analysis of Progression-free Survival

Subgroup	Trastuzumab Deruxtecan+ Pertuzumab	THP	Median Progression-free Survival (95% CI): Trastuzumab deruxtecan+ pertuzumab	Median Progression-free Survival (95% CI): THP	Hazard Ratio for Disease Progression or Death (95% CI)
	no. of events/no. of patients		*mo*		
All patients	118/383	172/387	40.7 (36.5–NC)	26.9 (21.8–NC)	0.56 (0.44–0.71)
Disease status					
Newly diagnosed	52/200	85/200	NC (36.5–NC)	31.2 (23.5–NC)	0.49 (0.35–0.70)
Recurrent	66/183	87/187	38.0 (26.9–NC)	22.5 (18.1–NC)	0.63 (0.46–0.87)
Hormone-receptor status					
Positive	65/207	87/209	38.0 (36.0–NC)	27.7 (22.4–NC)	0.61 (0.44–0.84)
Negative	53/176	85/178	40.7 (40.7–NC)	22.6 (17.3–32.7)	0.52 (0.37–0.73)
PIK3CA mutation status					
Detected	41/116	64/121	36.0 (29.7–NC)	18.1 (15.1–25.6)	0.52 (0.35–0.77)
Not detected	76/266	108/266	40.7 (38.0–NC)	32.7 (24.4–NC)	0.57 (0.43–0.77)
Brain metastasis at baseline					
Present	10/25	15/22	31.8 (18.5–NC)	9.5 (5.6–13.3)	0.30 (0.12–0.68)
Not present	108/358	157/365	40.7 (36.5–NC)	27.6 (22.6–NC)	0.58 (0.45–0.74)
Previous exposure to anti-HER2 therapies					
Yes	39/115	51/112	38.0 (26.9–NC)	21.5 (15.3–NC)	0.55 (0.36–0.83)
No	79/268	121/275	40.7 (36.5–NC)	27.6 (22.5–NC)	0.56 (0.42–0.74)
Previous exposure to pertuzumab					
Yes	5/31	12/26	40.8 (25.4–NC)	19.8 (7.5–NC)	NC
No	113/352	160/361	40.7 (36.0–NC)	27.4 (22.4–NC)	0.61 (0.48–0.77)
Previous exposure to endocrine therapy					
Yes	28/80	40/88	38.0 (25.4–NC)	23.7 (19.4–NC)	0.67 (0.41–1.09)
No	90/303	132/299	40.7 (36.5–NC)	27.4 (21.4–NC)	0.54 (0.41–0.70)
Previous exposure to anthracycline					
Yes	40/110	57/111	38.0 (25.7–NC)	19.4 (13.0–27.7)	0.50 (0.33–0.76)
No	78/273	115/276	40.7 (36.5–NC)	31.2 (23.3–NC)	0.58 (0.43–0.77)
Age at randomization					
<65 yr	90/315	139/315	40.7 (36.5–NC)	27.4 (22.4–NC)	0.50 (0.38–0.65)
≥65 yr	28/68	33/72	27.6 (14.9–NC)	21.5 (13.9–NC)	0.92 (0.55–1.51)
Geographic region					
Asia	62/188	87/191	40.7 (36.5–NC)	27.2 (21.5–NC)	0.60 (0.43–0.83)
Western Europe and North America	27/87	31/78	36.0 (30.6–NC)	31.2 (15.8–NC)	0.60 (0.35–1.01)
Rest of the world	29/108	54/118	NC (38.0–NC)	24.4 (14.8–NC)	0.48 (0.30–0.76)
Baseline ECOG performance-status score					
0	71/256	94/246	40.7 (36.5–NC)	NC (27.2–NC)	0.60 (0.44–0.81)
1	47/127	78/141	NC (26.3–NC)	19.2 (13.2–23.5)	0.54 (0.37–0.77)

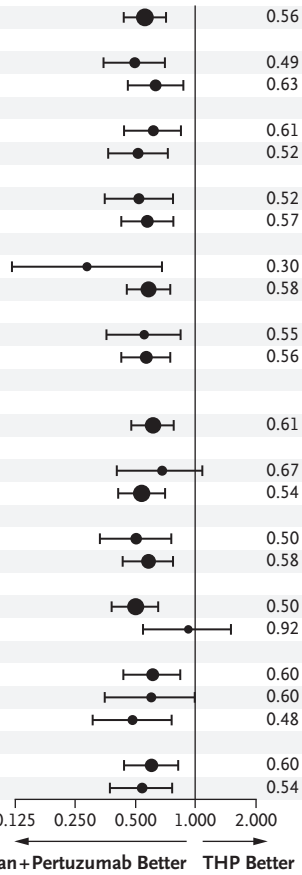


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Figure 1 (facing page). Progression-free Survival as Assessed by Blinded Independent Central Review (Full Analysis Population), Including Subgroup Analysis.
The full analysis population included the patients who underwent randomization. Panel A shows a Kaplan–Meier analysis of progression-free survival among patients assigned to receive trastuzumab deruxtecan plus pertuzumab and those assigned to receive a taxane, trastuzumab, and pertuzumab (a combination known as THP). Panel B shows a forest-plot analysis of progression-free survival according to subgroup. The sizes of the circles are proportional to the number of events. Eastern Cooperative Oncology Group (ECOG) performance-status scores are assessed on a 5-point scale, with higher scores indicating greater disability. HER2 denotes human epidermal growth factor receptor 2, and NC not calculable.

pertuzumab and 20.7 months (95% CI, 17.3 to 23.5) with THP (hazard ratio for disease progression or death, 0.49; 95% CI, 0.39 to 0.61) (Fig. S2). A progression-free survival benefit with trastuzumab deruxtecan plus pertuzumab was observed across most subgroups (Fig. 1B). Other data, including therapies after discontinuation of trial treatment, are provided in Table S2 and Figure S4.

Response to Treatment

Confirmed objective responses occurred in 326 patients (85.1%; 95% CI, 81.2 to 88.5) with trastuzumab deruxtecan plus pertuzumab and in 304 patients (78.6%; 95% CI, 74.1 to 82.5) with THP (odds ratio, 1.59; 95% CI, 1.09 to 2.32) (Table 2), with complete responses in 58 patients (15.1%) and 33 patients (8.5%), respectively. Investigator-assessed objective responses are shown in Table S1, and the best percentage change from baseline in target-lesion size is shown in Fig. S3. The median duration of response was 39.2 months (95% CI, 35.1 to not calculable) with trastuzumab deruxtecan plus pertuzumab and 26.4 months (95% CI, 22.3 to not calculable) with THP (Table 2).

Table 2. Antitumor Activity as Assessed by Blinded Independent Central Review (Full Analysis Population).*

Variable	Trastuzumab Deruxtecan plus Pertuzumab (N=383)	THP (N=387)
Confirmed objective response (95% CI) — %	85.1 (81.2–88.5)	78.6 (74.1–82.5)
Best confirmed response — no. (%)		
Complete response	58 (15.1)	33 (8.5)
Partial response	268 (70.0)	271 (70.0)
Stable disease	38 (9.9)	56 (14.5)
Progressive disease	13 (3.4)	12 (3.1)
Not evaluable	5 (1.3)	12 (3.1)
Median duration of response (95% CI) — mo	39.2 (35.1–NC)	26.4 (22.3–NC)

* Because of rounding and because patients with no evidence of disease are not included in this table, percentages may not total 100%. NC denotes not calculable.

SAFETY

Safety analyses included 381 patients who received trastuzumab deruxtecan plus pertuzumab and 382 patients who received THP. The median duration of treatment was 21.7 months (range, 0.3 to 44.5) among patients who received trastuzumab deruxtecan plus pertuzumab (with trastuzumab deruxtecan continued for a median of 20.0 months) and 16.9 months (range, 0.7 to 41.7) among patients who received THP (with a taxane given for a median of 4.4 to 5.5 months).

Overall safety was consistent with the known profiles of the individual treatments, with no new safety signals. Serious adverse events occurred in 103 of 381 patients (27.0%) with trastuzumab deruxtecan plus pertuzumab and in 96 of 382 patients (25.1%) with THP (Table S3); grade 3 or higher adverse events occurred in 242 patients (63.5%) and 238 patients (62.3%), respectively. Adverse events associated with dose reductions occurred in 175 patients (45.9%) with trastuzumab deruxtecan plus pertuzumab and in 76 patients (19.9%) with THP, and those associated with dose interruptions occurred in 262 (68.8%) and in 187 (49.0%), respectively. Adverse events leading to discontinuation of any trial treatment occurred in 79 patients (20.7%) with trastuzumab deruxtecan plus pertuzumab and in 108 patients (28.3%) with THP. The incidence of adverse events associated with death (grade 5) was 3.4% with trastuzumab deruxtecan plus pertuzumab and 0.8% with THP. Exposure-adjusted incidence rates of adverse events are shown in Table S4. The most common drug-related adverse event of any grade with trastuzumab deruxtecan plus pertuzumab was nausea; across both groups, diarrhea, fatigue, neutropenia, and alopecia were among the most common of such events (Table 3). The most common drug-related adverse events of grade 3 or higher were neutropenia, hypokalemia, and anemia with trastuzumab deruxtecan plus pertuzumab and neutropenia, leukopenia, and diarrhea with THP.

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Regarding adverse events of special interest, adjudicated drug-related interstitial lung disease or pneumonitis occurred in 46 patients (12.1%) with trastuzumab deruxtecan plus pertuzumab (grade ≤2 in 44 patients and grade 5 in 2 patients; the grade 5 events are described in Table S5) and in 4 patients (1.0%) with THP (grade ≤2 in all patients) (Table S6); the median time to onset was 210.5 days (range, 17 to 1028) and 294.5 days (range, 252 to 713), respectively. Left ventricular dysfunction was reported in 42 patients (11.0%) with trastuzumab deruxtecan plus pertuzumab (grade 4 event in 1 patient) and in 27 patients (7.1%) with THP.

DISCUSSION

In this prespecified interim analysis, trastuzumab deruxtecan plus pertuzumab resulted in a significant difference in progression-free survival as compared with THP, with benefit observed across most subgroups, in first-line HER2-positive advanced or metastatic breast cancer. The 44% relative reduction in the risk of disease progression or death indicates that patients may benefit from receiving trastuzumab deruxtecan plus pertuzumab as initial treatment for metastatic disease, an important point in the treatment pathway after which an estimated 25 to 30% of patients currently may not receive second-line therapy.[23,24] Given the current interim follow-up, the median values for progression-free survival may evolve further, but the fact that 70.1% of patients receiving trastuzumab deruxtecan plus pertuzumab were estimated to remain free of progression or death at 24 months (vs. 52.1% with THP) is of clinical relevance and indicative of a sustained benefit in the first-line context.

A high incidence of confirmed objective response and a median duration of response exceeding 39 months suggest durability of effect with trastuzumab deruxtecan plus pertuzumab, with the incidence of complete response nearly twice that with THP. Future analyses will characterize additional outcomes such as intracranial efficacy. Initial subgroup data indicate that the progression-free survival benefit of trastuzumab deruxtecan plus pertuzumab as compared with THP was consistently observed regardless of the presence or absence of brain metastases at baseline (albeit that relatively few patients with brain metastases at baseline were enrolled). Data from the second-line context (DESTINY-Breast12 trial) indicate that trastuzumab deruxtecan has substantial and durable activity (12-month progression-free survival, 61.6%), including intracranial activity (12-month central nervous system [CNS] progression–free survival, 58.9%) in patients with active or stable brain metastases at baseline.[25] CNS efficacy will probably be an important consideration in the potential first-line use of trastuzumab deruxtecan.

The efficacy of THP in this context was established in the CLEOPATRA trial, which showed significant differences in progression-free and overall survival as compared with trastuzumab plus docetaxel.[9,10] After adoption of THP, treatment strategies have continued to evolve, particularly in hormone receptor–positive, HER2-positive metastatic breast cancer. In the PERTAIN trial, adding pertuzumab to endocrine therapy (aromatase inhibitor) and trastuzumab plus or minus chemotherapy led to prolonged progression-free survival among patients with hormone receptor–positive disease.[26] In the DESTINY-Breast09 trial, approximately 38% of patients in the THP group who had hormone receptor–positive tumors received concurrent endocrine therapy, and this situation could have contributed to the longer median progression-free survival with THP than that seen in the CLEOPATRA trial.[9,10] More recently, the PATINA trial showed a benefit in adding palbociclib to standard dual HER2-directed therapy plus endocrine therapy in patients who had at least stable disease after taxane therapy (hazard ratio for investigator-assessed disease progression or death, 0.74; P=0.0074).[27] The design of the PATINA trial is notably different from that of the DESTINY-Breast09 trial, in which an internationally representative patient population unrestricted by hormone-receptor status was randomly assigned to treatment that was planned to continue until disease progression occurred. Although patients with hormone receptor–positive status could receive concurrent endocrine therapy in the DESTINY-Breast09 trial, these patients were limited in number, with an imbalance between treatment groups. Of note, the progression-free survival benefit with trastuzumab deruxtecan plus pertuzumab was consistent regardless of hormone-receptor status.

At the interim analysis, the median progression-free survival with trastuzumab deruxtecan plus pertuzumab was 40.7 months as assessed by

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Table 3. Drug-Related Adverse Events (Safety Analysis Population).*

Event	Trastuzumab Deruxtecan plus Pertuzumab (N=381)		THP (N=382)	
	All Grades	Grade ≥3	All Grades	Grade ≥3
	number of patients (percent)			
Any adverse event	373 (97.9)	209 (54.9)	369 (96.6)	200 (52.4)
Nausea	271 (71.1)	19 (5.0)	110 (28.8)	1 (0.3)
Diarrhea	213 (55.9)	26 (6.8)	207 (54.2)	20 (5.2)
Neutropenia†	186 (48.8)	91 (23.9)	170 (44.5)	127 (33.2)
Fatigue‡	184 (48.3)	30 (7.9)	132 (34.6)	8 (2.1)
Alopecia§	176 (46.2)	0	191 (50.0)	2 (0.5)
Vomiting	160 (42.0)	9 (2.4)	51 (13.4)	2 (0.5)
Transaminases increased¶	137 (36.0)	17 (4.5)	72 (18.8)	8 (2.1)
Anemia‖	135 (35.4)	32 (8.4)	149 (39.0)	14 (3.7)
Leukopenia**	112 (29.4)	17 (4.5)	117 (30.6)	67 (17.5)
Decreased appetite	109 (28.6)	9 (2.4)	59 (15.4)	3 (0.8)
Weight decreased	91 (23.9)	10 (2.6)	26 (6.8)	1 (0.3)
Thrombocytopenia††	89 (23.4)	24 (6.3)	17 (4.5)	3 (0.8)
Constipation	85 (22.3)	1 (0.3)	26 (6.8)	0
Hypokalemia	82 (21.5)	39 (10.2)	24 (6.3)	6 (1.6)
Peripheral sensory neuropathy‡‡	43 (11.3)	0	109 (28.5)	4 (1.0)

* The safety population included the patients who received at least one dose of a trial drug. Shown are adverse events that occurred in at least 20% of the patients in either trial group and were determined by the investigator to be related to a trial drug. They are listed in decreasing frequency among patients receiving trastuzumab deruxtecan plus pertuzumab and included those that emerged or worsened on or after the date of the first dose of a trial drug up to and including 47 days after the last dose of a trial drug or before the initiation of the first subsequent cancer therapy (whichever occurred first). Adverse events were coded with the use of the preferred terms of the *Medical Dictionary for Regulatory Activities*, version 27.1, and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.0.

† This category includes the preferred terms neutrophil count decreased and neutropenia.

‡ This category includes the preferred terms fatigue, asthenia, malaise, and lethargy.

§ Given that alopecia severity can be classified no higher than grade 2, it is assumed that the two patients with a recorded grade 3 or higher event represents a misclassification.

¶ This category includes the preferred terms transaminases increased, aspartate aminotransferase increased, alanine aminotransferase increased, γ-glutamyl transferase increased, liver function test abnormal, hepatic function abnormal, and liver function test increased.

‖ This category includes the preferred terms hemoglobin decreased, red-cell count decreased, anemia, and hematocrit decreased.

** This category includes the preferred terms white-cell count decreased and leukopenia.

†† This category includes the preferred terms platelet count decreased and thrombocytopenia.

‡‡ This category includes the preferred terms neuropathy peripheral, peripheral sensory neuropathy, and polyneuropathy.

blinded independent central review and as assessed by investigators; in the THP group, the value was 26.9 months as assessed by blinded independent central review but 20.7 months as assessed by investigators. After progression was determined by investigators, additional imaging assessments were performed to facilitate blinded independent central review with respect to the primary end point. Although every effort was made to collect these additional tumor scans (according to the protocol), there remained patients in the trial for whom blinded review did not subsequently confirm investigator-assessed progression, and data for these patients were censored for progression-free survival by blinded review. Given the higher incidence of disease progression or death with THP than with trastuzumab deruxtecan plus pertuzumab, the THP group contained more

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of these patients with censored data; consequently, the discrepancy between blindly assessed and investigator-assessed progression-free survival was more pronounced for THP than for trastuzumab deruxtecan plus pertuzumab. Regardless, trastuzumab deruxtecan plus pertuzumab showed a benefit in progression-free survival as assessed by either blinded independent central review or by investigators.

Trastuzumab deruxtecan has shown efficacy in later treatment lines in the post–T-DM1 context (DESTINY-Breast02 trial)[28] and post-THP context (DESTINY-Breast03 trial).[15] Although data for trastuzumab deruxtecan plus placebo in the DESTINY-Breast09 trial remain blinded until the final analysis of progression-free survival, relevant data in first-line metastatic breast cancer have been reported from an interim analysis of the DESTINY-Breast07 trial, in which 12-month progression-free survival for trastuzumab deruxtecan monotherapy and trastuzumab deruxtecan plus pertuzumab (80.8% and 89.4%, respectively) suggest that trastuzumab deruxtecan monotherapy may also provide benefit.[29] Subsequent results from both trials are needed to further validate this.

Trastuzumab deruxtecan plus pertuzumab had similar safety as compared with THP, with approximately 60% of patients in both groups having adverse events of grade 3 or higher and approximately one quarter of patients having serious adverse events. For trastuzumab deruxtecan, these findings are consistent with what has been observed in previous studies involving patients with HER2-positive metastatic breast cancer[15,28] and other forms of breast cancer,[30,31] with gastrointestinal disorders among the most common adverse events. Although not mandatory, antiemetic prophylaxis was recommended, and the lower incidence of grade 3 nausea and vomiting than that in the DESTINY-Breast03 trial[15] may reflect growing clinical experience of managing trastuzumab deruxtecan–related gastrointestinal effects. Adverse events involving interstitial lung disease or pneumonitis with trastuzumab deruxtecan plus pertuzumab were predominantly grade 1 or 2, although two events were associated with death; interstitial lung disease or pneumonitis remains an important adverse event associated with trastuzumab deruxtecan for which patients should be monitored appropriately.[21,22]

With 1157 patients in this international trial, results are expected to be widely relevant to clinical practice worldwide. Current data are sufficiently mature to show a significant difference between treatments, although median progression-free survival for trastuzumab deruxtecan plus pertuzumab (40.7 months) was largely driven by a small number of patients who have been subject to relatively long follow-up. General limitations of the trial include the partial open-label design and, despite its international reach, the underrepresentation of certain groups, notably Black patients, which could affect the generalizability of the findings (Table S7).

In this trial, trastuzumab deruxtecan plus pertuzumab showed a significant benefit in progression-free survival as compared with THP in advanced or metastatic HER2-positive breast cancer, a benefit that was consistently observed across most patient subgroups. No new safety signals were identified.

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A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

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